

Question - 1: What Do You Mean by Addressing Modes? List the different types of addressing modes.

In a computer, when a program runs, the CPU needs to fetch data from memory or registers to process it. Addressing modes are the different ways a CPU can locate and access this data. They tell the processor where to find the operand (data) required for an instruction.

Think of it like finding a contact on your phone. You can dial a number directly (immediate mode), search by name (direct mode), or use a shortcut (indexed mode). Similarly, addressing modes help the CPU access data efficiently.

Different addressing modes are used based on how flexible and efficient the program needs to be. Some modes make instructions shorter, while others allow access to large amounts of memory.

Different Types of Addressing Modes

1. Immediate Addressing Mode

- The actual data (operand) is written inside the instruction.
- **Example:** MOV A, 10 (Moves the number 10 directly into register A).

2. Register Addressing Mode

- The operand is stored in a register, so the instruction just needs to refer to it.
- **Example:** ADD A, B (Adds the values of registers A and B).

3. Direct Addressing Mode

- The memory location where the operand is stored is given in the instruction.
- **Example:** MOV A, [3000] (Takes data from memory address 3000 and stores it in A).

4. Indirect Addressing Mode

- Instead of directly mentioning the memory address, a register holds the address where data is stored.
- **Example:** MOV A, [R1] (R1 contains an address, and the value at that address is moved into A).

5. Indexed Addressing Mode

- Uses a base address plus an index value to fetch data, mostly used in arrays.
- **Example:** Accessing the 5th element in an array stored in memory.

6. Relative Addressing Mode

- Used mainly for jump instructions, where the address is given relative to the current instruction.

Addressing modes are essential in programming and CPU design because they help in efficient memory access, shorter instructions, and better performance.

Question - 2: What are the functions of the control unit? Explain.

Functions of the Control Unit

The Control Unit (CU) is one of the most important parts of a computer's Central Processing Unit (CPU). It acts like the brain inside the brain, directing the flow of data and instructions within the computer. Without the control unit, the CPU wouldn't know what to do next, and the whole system would be a mess.

The control unit's job is to fetch, decode, and execute instructions, making sure everything in the CPU and other components work together properly. It does not actually process data like the ALU (Arithmetic Logic Unit), but it ensures that instructions are carried out in the correct order.

Main Functions of the Control Unit

1. Fetching Instructions

The first thing the control unit does is fetch instructions from memory (RAM). These instructions tell the CPU what task it needs to perform. The CU reads these instructions and gets them ready for processing. For example, when you press a key on your keyboard, the CU fetches the instruction for displaying that key on the screen.

2. Decoding Instructions

Once the instruction is fetched, the CU decodes it. This means it breaks down the instruction into a form that the CPU can understand. It determines what operation needs to be performed, whether it's an arithmetic calculation, data movement, or logic operation.

3. Controlling Data Flow

The CU manages the movement of data inside the CPU and between different parts of the computer. It makes sure that data goes to the right place at the right time. If an instruction needs data from memory, the CU tells the Memory Unit to send it. If an instruction requires a calculation, it signals the ALU to perform the operation.

4. Managing Execution of Instructions

After decoding, the CU sends control signals to different parts of the computer to execute the instruction. It tells the ALU to perform arithmetic operations, the memory to store or retrieve data, and the output devices to display the result. For example, when playing a video, the CU coordinates between the graphics card, processor, and speakers to ensure smooth playback.

5. Handling Interrupts

An interrupt happens when a device or software requests the CPU's attention. The CU detects and handles interrupts, deciding if it should pause the current task and deal with the new

request first. For example, if you're typing and suddenly receive a WhatsApp notification, the CU helps the processor switch tasks temporarily.

The Control Unit is like a traffic cop inside the CPU, making sure every part of the computer works in sync. It fetches, decodes, and executes instructions while managing data flow and handling tasks efficiently. Without it, a computer would be useless, as nothing would get processed properly.

Question - 3: Explain the concept of memory interleaving.

Concept of Memory Interleaving

Memory interleaving is a technique used in computer memory systems to improve the speed and efficiency of data access. It helps the CPU access data faster by organizing memory in a way that reduces waiting time.

Imagine you're in a school library, and instead of having all books stored in one long row, they are divided into multiple smaller shelves. This way, many students can access books at the same time without waiting. Memory interleaving works in a similar way by splitting memory into smaller banks that can be accessed simultaneously.

Why Do We Need Memory Interleaving?

In a computer, the CPU is much faster than the memory. When the CPU requests data from memory, it often has to wait because memory takes time to fetch data. This creates a bottleneck that slows down performance. Memory interleaving helps solve this issue by allowing multiple memory banks to work together, so data retrieval happens faster and more efficiently.

Without interleaving, the CPU would have to wait longer for data, leading to delays in executing programs.

How Memory Interleaving Works

Memory interleaving divides the system memory into multiple banks and arranges them in a way that allows the CPU to fetch data from different banks at the same time.

There are two main types of memory interleaving:

- Low-order interleaving
 - Memory addresses are assigned sequentially across different banks.
 - Example: Address 0 goes to Bank 0, Address 1 to Bank 1, Address 2 to Bank 2, and so on.
 - This means the CPU can access multiple memory banks simultaneously, reducing wait time.

- High-order interleaving
 - Consecutive memory addresses are stored in the same bank before moving to the next bank.
 - Example: Addresses 0-3 go to Bank 0, addresses 4-7 go to Bank 1, etc.
 - This is useful when memory needs to be accessed in large chunks.

Advantages of Memory Interleaving

- Faster Data Access – Since multiple memory banks are accessed at the same time, data retrieval is quicker.
- Reduced CPU Waiting Time – The CPU doesn't have to wait long for memory responses, making the system more efficient.
- Better Performance for Multi-Core Processors – Modern processors can access memory banks independently, improving multitasking and gaming performance.
- Efficient Utilization of Memory – Memory banks are used more effectively, avoiding bottlenecks.

Memory interleaving is a smart way to speed up data access and improve system performance. By breaking memory into multiple banks and accessing them in parallel, computers can handle tasks much faster. It is especially useful in high-performance computing, gaming PCs, and servers, where speed matters the most.

Question - 4: Write a note on DVD.

A Note on DVD

A DVD (Digital Versatile Disc) is an optical storage device used to store data like movies, music, software, and other digital content. It looks just like a CD (Compact Disc) but has a much higher storage capacity. DVDs became very popular in the late 1990s and early 2000s as a way to store and watch high-quality videos.

Storage Capacity & Types of DVDs

One of the biggest advantages of DVDs over CDs is that they can store more data. A standard CD can hold about 700 MB, while a DVD can hold 4.7 GB or more, depending on its type.

There are different types of DVDs based on their storage capacity and functionality:

1. DVD-5 (Single-layer, single-sided) – Stores **4.7 GB** of data.
2. DVD-9 (Dual-layer, single-sided) – Stores **8.5 GB** of data.
3. DVD-10 (Single-layer, double-sided) – Stores **9.4 GB** of data.
4. DVD-18 (Dual-layer, double-sided) – Stores **17 GB** of data.

Some DVDs are read-only (DVD-ROM), meaning they come pre-recorded, like movie DVDs. Others are rewritable (DVD-RW, DVD+RW), allowing users to record and erase data multiple times.

How DVDs Work

DVDs store data using tiny pits and lands on their surface, which are read by a laser beam inside a DVD player or drive. The laser reads the pattern of pits and converts it into digital data that can be played as a video, audio, or computer file.

DVDs use a smaller laser wavelength (650 nm) compared to CDs (780 nm), allowing them to store more data in the same space. This is why DVDs have better picture and sound quality than CDs.

Advantages of DVDs

- **High Storage Capacity** – A DVD can store movies, games, and software that wouldn't fit on a CD.
- **Better Quality** – DVDs offer higher resolution video and clearer sound, making them ideal for movies and games.
- **Portability** – DVDs are small, lightweight, and easy to carry.
- **Cost-Effective** – Even though new technologies like USB and cloud storage exist, DVDs are still an affordable option for backups and distribution.

Disadvantages of DVDs

- **Easily Scratched** – The surface of a DVD can get scratched easily, making it unreadable.
- **Requires a DVD Drive** – Many modern laptops and devices no longer come with DVD drives.
- **Slower Access** – Compared to USB drives or SSDs, DVDs take longer to read and write data.

DVDs were once the best way to store and watch high-quality movies, install software, and back up files. While they have become less common today due to USB drives, streaming, and cloud storage, they are still used in many places, especially for movies, gaming consoles, and archival storage.

Question - 5: Describe briefly PCI bus and its importance.

Brief Description of PCI Bus and Its Importance

The PCI (Peripheral Component Interconnect) bus is a hardware interface used in computers to connect expansion cards like graphics cards, sound cards, network adapters, and storage controllers to the motherboard. It was introduced in the early 1990s and became one of the most commonly used expansion bus standards in personal computers.

A bus in a computer is like a highway that allows different components to communicate with each other. The PCI bus helps in transferring data quickly and efficiently between the CPU, RAM, and connected devices.

Structure and Working of PCI Bus

The PCI bus is a parallel communication bus, meaning multiple bits of data are transferred simultaneously. It uses a shared bus structure, where all connected devices share the same data path. The PCI slots on the motherboard allow users to install additional hardware components to improve the functionality of their computer.

PCI operates at different speeds and versions, such as:

- PCI 1.0 – 32-bit, 33 MHz speed, offering 133 MB/s transfer rate.
- PCI 2.0 and 2.1 – Improved speed up to 266 MB/s.
- PCI 3.0 – Allowed 64-bit operation, doubling the transfer speed.

Later, PCI evolved into PCI-X (PCI Extended) and PCI Express (PCIe), which replaced traditional PCI with faster speeds and better performance.

Importance of PCI Bus

1. Expanding System Capabilities

PCI allows users to add new hardware components to their computer. If someone needs a better graphics card for gaming or video editing, they can install one into a PCI slot without replacing the whole motherboard.

2. Faster Data Transfer

Compared to older buses like ISA (Industry Standard Architecture), PCI provides much faster data transfer rates, improving the overall system performance. It supports high-speed components like network cards and SSD controllers.

3. Compatibility

PCI became a standard in the computer industry, meaning most expansion cards are designed to work with PCI slots, making it easier to upgrade hardware.

4. Multipurpose Use

It is used for various components like:

Graphics Cards – Enhances gaming and video editing.

Sound Cards – Improves audio quality for music production.

Network Cards – Enables fast internet and LAN connectivity.

Storage Controllers – Connects SSDs and RAID arrays for better data handling.

The PCI bus played a major role in improving the flexibility and performance of computers by allowing easy hardware expansion. Though modern computers now use PCI Express (PCIe) for even faster speeds, the PCI bus remains a foundation for how expansion slots work in a computer.

Question - 6: Differentiate between multiprocessing and multiprogramming.

Difference Between Multiprocessing and Multiprogramming

Computers handle multiple tasks at the same time using different techniques. Two important ones are **multiprocessing** and **multiprogramming**. While both improve system performance, they work in different ways.

What is Multiprocessing?

Multiprocessing is when a computer has two or more processors (CPUs) working together to perform multiple tasks at the same time. Each processor can handle different tasks independently, making the system faster and more efficient.

How It Works

- In a dual-core or quad-core processor, each core can execute a different process at the same time.
- If you're watching a YouTube video while playing a game, one processor might handle the video, while another manages the game.

Advantages of Multiprocessing

Faster Performance – Multiple CPUs can divide tasks, reducing the processing time.

Better for Heavy Tasks – Used in servers, gaming, AI, and scientific research.

More Reliable – If one processor fails, others can still function.

Example of Multiprocessing

- High-end computers, servers, and supercomputers use **multiprocessing** for handling large amounts of data.
- **Intel Core i7 or AMD Ryzen processors** have multiple cores that help in multiprocessing.

What is Multiprogramming?

Multiprogramming is when a computer runs multiple programs at the same time, but only one program is actively executing at a time. The CPU switches between programs so that users feel like multiple tasks are happening at once.

How It Works

- The CPU executes one program while keeping others in the waiting state.
- When one program needs to wait for input/output (I/O) operations, the CPU switches to another program to keep the system busy.

Advantages of Multiprogramming

Efficient CPU Usage – The CPU is never idle; it switches between tasks.

Faster Response Time – Users can run multiple applications without delays.

Good for General Users – Works well in normal desktops and office environments.

Example of Multiprogramming

- When you open a web browser, a text editor, and listen to music at the same time, your operating system switches between them, but only one is using the CPU at any moment.

Key Differences Between Multiprocessing and Multiprogramming

Feature	Multiprocessing	Multiprogramming
Number of CPUs	More than one CPU	Only one CPU
Execution	Multiple tasks run simultaneously	Tasks run one at a time, but CPU switches between them

Efficiency	Faster because tasks are divided among CPUs	Efficient use of CPU time but slower than multiprocessing
Usage	Used in servers, gaming, AI, high-performance computing	Used in general-purpose computers, desktops, and office systems
Example	Supercomputers, modern multi-core processors	Running multiple apps on a single-core PC

Multiprocessing is about multiple CPUs working at the same time, while multiprogramming is about one CPU switching between different tasks. Multiprocessing is faster and more powerful, but multiprogramming helps in better CPU utilization for normal computers. Both techniques are essential for improving computer performance and efficiency